

00. Start Here - Absolute Beginner

No previous computer programming knowledge required

This guide explains the words and actions used by every later PDF. Complete it before Web Foundations.

ابدأ من هنا. لا تحتاج إلى أي معرفة سابقة بالبرمجة. ادرس هذا الدليل قبل بقية الكتب.

Learning rule

Type every example. Do not paste it. A typing mistake is useful practice because learning to read errors is part of programming.

Lesson 1 - What programming is

Explanation

A program is a precise list of instructions that a computer executes. Source code is human-readable text. An interpreter or compiler translates it into work the computer can perform.

Example

input -> instructions -> output

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Describe the input and output of a calculator.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 2 - Files, folders, and extensions

Explanation

A file stores content and has a name. A folder organises files. The extension after the final dot suggests the format: .py for Python, .html for HTML, .ts for TypeScript, .json for structured text, and .pdf for documents.

Example

```
Learning/  
  hello.py  
  index.html  
  data/  
    summary.json
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Create this folder structure in a safe learning folder.

اكتب المثال بنفسك، ثم شغله وغير شيئا واحدا. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 3 - Installing and using VS Code

Explanation

VS Code is a text editor for code. Open a folder, use the Explorer to create files, use tabs to edit them, save with Ctrl+S, and read highlighted syntax and problems. Extensions add language support but do not replace the language runtime.

Example

File -> Open Folder
Explorer -> New File
Terminal -> New Terminal
Ctrl+S -> Save

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Open learning-examples/00-start-here in VS Code.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 4 - PowerShell and the terminal

Explanation

A terminal lets you type commands. PowerShell shows a prompt. The current directory decides which files a command sees. Use `pwd` to show it, `Get-ChildItem` to list items, `cd` to change folder, and `clear` to clear the screen.

Example

```
pwd
Get-ChildItem
cd .\learning-examples\00-start-here
Get-ChildItem
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Navigate to the practice folder and list its files.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 5 - Commands, arguments, and errors

Explanation

The first word names a command; later words are arguments. Quoted paths preserve spaces. Exit code zero normally means success. Error output often names the missing command, file, package, or source line.

Example

```
python hello.py
# command: python
# argument: hello.py
cd "C:\A Folder With Spaces"
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Run one valid command and one deliberately misspelled command; compare messages.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 6 - Installing Python

Explanation

Python runs backend and data examples. During installation select the option that adds Python to PATH. Verify with `python --version`. `pip` installs packages, while `python -m pip` ensures `pip` belongs to the selected Python.

Example

```
python --version
python -m pip --version
python -c "print('Python works')"
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Run the three verification commands.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 7 - Your first Python program

Explanation

Create hello.py, type a print call, save, then run it from the same folder. Quotes create text. Parentheses pass the text to print. The terminal shows the program's output.

Example

```
# hello.py
print('Hello! I ran my first Python program.')
# PowerShell
python hello.py
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Change the message and run the file again.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 8 - Installing Node.js and npm

Explanation

Node.js runs JavaScript outside the browser and supplies npm, the package manager used by frontend projects. Verify both commands. package.json describes scripts and dependencies for one project.

Example

```
node --version
npm --version
npm init -y
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Verify Node and npm; do not install packages globally unless a guide requires it.

اكتب المثال بنفسك، ثم شغله وغير شيئا واحدا. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 9 - Your first browser page

Explanation

A browser opens an HTML file. HTML tags describe content. Save index.html, double-click it, then refresh after changes. Browser developer tools open with F12 and show elements, console messages, and network activity.

Example

```
<!doctype html>
<html>
<body>
  <h1>My first page</h1>
  <p>Hello, browser!</p>
</body>
</html>
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Change the heading, save, and refresh the browser.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 10 - Dependencies and package managers

Explanation

A dependency is code your project uses. `npm install` reads `package.json` for JavaScript. `pip install -r requirements.txt` reads Python requirements. Dependencies belong to the project and should use known compatible versions.

Example

```
npm install
python -m pip install -r requirements.txt
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Find `package.json` and `requirements.txt` in the learning examples.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 11 - Python virtual environments

Explanation

A virtual environment gives one Python project its own installed packages. Create .venv, activate it, install requirements, and deactivate when finished. The prompt normally shows (.venv) while active.

Example

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
deactivate
```

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Create and activate a virtual environment in a disposable practice folder.

اكتب المثال بنفسك، ثم شغله وغير شيئا واحدا. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Lesson 12 - How to learn and debug

Explanation

Do not memorise everything. Type small examples, predict results, change one value, and explain the result. When an error occurs, read it fully, locate the first relevant file and line, inspect the value, and change one thing at a time.

Example

Expected: 5
Actual: TypeError on line 4
Cause: text was added to a number
Fix: convert or validate input
Test: normal, empty, invalid

Do it step by step

1. Read the example.
2. Type it exactly.
3. Save the file if applicable.
4. Run the command.
5. Compare what happened with the explanation.
6. Change one small value and repeat.

Try it yourself

Start an error notebook using these five labels.

اكتب المثال بنفسك، ثم شغله وغير شيئاً واحداً. إذا ظهر خطأ، اقرأ الرسالة كاملة.

Ready for Guide 01

Before continuing, you should be able to create a folder and file, open a folder in VS Code, navigate with PowerShell, run a Python file, open an HTML page, verify Python/Node/npm, and explain the difference between source code, a command, a dependency, input, and output.

If any item is unclear, repeat the matching lesson. Then continue to 01 - Web Foundations.